KANO ELECTRICITY DISTRIBUTION COMPANY

May 2026 11kV Feeder Performance

Management Report

Prepared from the 11kV feeder performance monitoring output for May 2026

Average Supply	Average Load	Energy Delivered	Interruptions
7.32 hrs/day	**0.54 MW**	**57,029.86 MWh**	**5,781 / 5,783**

May 2026

Contents

1. Executive Management Summary

The May 2026 11kV feeder performance review shows a material improvement in average hours of supply and energy delivered when compared with the previous reporting period. Average hours of supply increased to 7.32 hours per feeder per day, compared with a previous-period average of 5.01 hours, representing a 46.2 percent improvement. Energy delivered also improved significantly, with the daily energy trend showing an average of 2,241.58 MWh, compared with 1,372.22 MWh in the previous period, representing a 63.4 percent increase.

Despite this improvement, the overall service position remains weak. The average supply level of 7.32 hours per day is still low when assessed against the expected service levels for the different service bands. Band A recorded a relatively strong average supply of 19.83 hours per day, but Bands B, C and D recorded weak averages of 5.98 hours, 4.49 hours and 3.05 hours respectively. Band E recorded 6.94 hours, but this is based on only one feeder.

The reliability position also requires urgent attention. The system recorded 64,635 cumulative interruption hours, an average interruption duration of 16.68 hours, and an average local fault turnaround time of 7.26 hours. This indicates that interruptions remain both frequent and prolonged, and that improved supply availability has not yet resolved the underlying reliability challenge.

The interruption breakdown shows a major data-quality and operational-diagnosis issue. The largest interruption category is N/A, with 2,768 interruptions and 23,077.6 hours. This is followed by L/S, with 2,408 interruptions and 21,691.4 hours. Together, these categories dominate the interruption profile. The high level of uncategorised interruptions reduces management's ability to diagnose root causes and take targeted corrective action.

At feeder level, there is a wide gap between strong-performing and weak-performing feeders. Several Band A feeders recorded strong supply performance above 20 hours per day, including Industrial, Ceramic, Funtua Water Works, Yusuf Road, NBC, Dala Foods, Maimalari and Bompai. However, several feeders across Bands B, C and D recorded very low supply, with some feeders showing data combinations that require validation, such as zero supply hours but positive energy delivered.

Management priority for the next reporting cycle

Protect the improvement in supply hours, reduce the interruption burden, clean up interruption classification, validate feeder-level data anomalies, and implement a focused service-band recovery plan for Bands B, C and D.

2. Headline KPI Dashboard

KPI	May 2026 Position	Previous / Comparator	Movement	Management Interpretation	Status
Average hours of supply	7.32 hrs/day	5.01 hrs/day	+46.2%	Supply improved significantly, but remains low at the overall company level.	Improved but still weak
Average load	0.54 MW	0.55 MW	-2.1%	Load declined slightly despite improved supply hours.	Watch
Energy delivered	57,029.86 MWh total reported	Previous daily average of 1,372.22 MWh	+63.4% based on daily trend	Energy delivery improved materially, likely driven by longer supply availability rather than higher average load.	Positive, subject to validation
Total interruptions	5,781 in technical overview	5,783 in state and band summaries	Difference of 2	Interruption totals need to be reconciled across report sections.	Data validation required
Cumulative interruption hours	64,635 hrs	Not stated	Not stated	Overall interruption exposure remains high.	High concern
Average interruption duration	16.68 hrs	Not stated	Not stated	Interruptions are prolonged and continue to affect service reliability.	High concern
Local fault turnaround time	7.26 hrs	Not stated	Not stated	Local fault restoration time remains a key operational performance issue.	High concern
Total monitored feeders	125 feeders	Not stated	Not stated	The report provides a broad feeder-level monitoring base for management accountability.	Useful

Key message

Supply availability improved, but reliability, service-band performance and data quality still require management attention.

3. Overall Performance Interpretation

The May 2026 performance shows a positive supply-side movement, but not yet a strong service-delivery outcome. The company delivered more hours of supply and more energy compared with the previous reporting period, but the service-band summary shows that most bands remain significantly below the expected level of service.

The increase in energy delivered appears to have been driven more by improved supply availability than by load growth. This is because average load declined slightly from 0.55 MW to 0.54 MW, while average hours of supply increased by 46.2 percent. This means the company was able to deliver more energy largely because feeders were available for longer periods, not because average loading improved.

This distinction is important for management. If supply hours improved but load declined, the company should assess whether the decline in load reflects actual customer demand, feeder switching, suppressed demand, metering gaps, customer non-payment, system constraints, or data estimation issues.

There is also a need to clarify the basis for energy reporting. The technical overview reports total delivered energy of 57,029.86 MWh, while the energy delivered trend page shows a daily average of 2,241.58 MWh. If the daily average is multiplied by 31 days, it implies a higher monthly total than the technical overview. This does not automatically mean the report is wrong, but it means the calculation basis should be clarified. The report should explain whether the two figures are based on the same feeder population, same reporting days, same metering logic, or whether one includes system estimates while the other is based on validated totals.

4. Reliability and Interruption Management Review

Reliability remains the most critical operational issue in the May 2026 report. The system recorded 64,635 cumulative interruption hours, with an average interruption duration of 16.68 hours. This suggests that outages were not only frequent, but also long enough to materially affect customer service experience and feeder-level performance.

The interruption breakdown shows that the highest interruption categories are N/A and L/S. This creates two management concerns. First, the high N/A category indicates that a significant share of interruptions is not being properly classified. Second, the high L/S category suggests that load-supply-related interruptions are a dominant operational issue and should be separated into more useful subcategories.

4.1 Interruption Breakdown

Interruption Category	Count	Total Hours	Average Duration	Management Interpretation
N/A	2,768	23,077.6 hrs	8.3 hrs	Largest category. This is a major data-quality issue because the cause is not properly classified.
L/S	2,408	21,691.4 hrs	9.0 hrs	Major driver of interruptions. Requires clearer separation between load shedding, supply constraint and operational switching.
L/S GS	155	644.2 hrs	4.2 hrs	Requires clearer operational definition.
L/F	82	728.5 hrs	8.9 hrs	Local fault-related interruptions require stronger technical response tracking.
O/S	75	619.9 hrs	8.3 hrs	Should be reviewed to clarify outage source and responsible control point.
TCN	74	152.7 hrs	2.1 hrs	Transmission-related outages are lower in total duration but should remain separately monitored.
E/F	74	509.2 hrs	6.9 hrs	Earth-fault events require technical follow-up and feeder-specific analysis.
Permit	42	96.8 hrs	2.3 hrs	Planned outages should be monitored for scheduling and customer communication.
Fault	17	200.3 hrs	11.8 hrs	Smaller count but high average duration. Requires investigation.
OC & E/F	2	34.9 hrs	17.5 hrs	Low count but very high average duration. Requires technical review.

4.2 Reliability Management Implications

Issue	Implication	Required Management Response
High cumulative interruption hours	Customers experienced extended supply disruption across the monitored feeder base.	Prioritise feeders with the highest interruption duration for operational review.
High average interruption duration	Restoration times are long and may be worsening customer dissatisfaction.	Track restoration cycle by fault report time, dispatch time, arrival time, restoration time and closure time.
Large N/A category	Management cannot fully diagnose the cause of interruptions.	Reclassify N/A interruptions and enforce mandatory interruption-cause coding.
High L/S interruptions	Supply constraints are a major contributor to poor service delivery.	Separate controllable and externally driven L/S events.
Local fault turnaround time of 7.26 hours	Local fault response still requires improvement.	Create feeder-level fault response scorecards and escalation thresholds.

5. Data Quality and Validation Review

The report is useful, but several data-quality issues need to be addressed before it can fully function as a management report.

5.1 Key Data Issues Identified

Data Issue	Observation	Why It Matters	Recommended Correction
Interruption totals do not fully reconcile	Technical overview shows 5,781 interruptions, while state and service-band summaries show 5,783.	A small discrepancy can still weaken confidence in the report.	Reconcile interruption totals across all report sections before publication.
Energy reporting basis requires clarification	Total energy delivered is shown as 57,029.86 MWh, while daily trend average is 2,241.58 MWh.	Different energy figures may confuse management if the calculation basis is not explained.	Add methodology note explaining total energy, daily average, metered energy and system-estimated energy.
Large N/A interruption category	N/A accounts for 2,768 interruption events.	The largest category does not support root-cause decision-making.	Reduce N/A classification through mandatory coding and post-event review.
Zero supply but positive energy	Some feeders show zero or near-zero supply hours but positive energy delivered.	This may indicate estimation, missing supply data, missing load data or data integration issues.	Flag these feeders as exceptions and validate meter/system data.
Service-band underperformance	Several bands are far below expected service levels.	This has customer-service, regulatory and commercial implications.	Add actual vs expected service-band compliance analysis.

5.2 Feeder-Level Data Exceptions Requiring Validation

Feeder	Band	Reported Supply	Peak Load	Energy Delivered	Issue
Hausawa	B	0.00 hrs	0.00 MW	533.91 MWh	Zero supply and zero load but positive energy.
Low Cost	B	0.00 hrs	0.00 MW	253.78 MWh	Zero supply and zero load but positive energy.
Sani Abacha Way	B	0.06 hrs	0.75 MW	252.46 MWh	Near-zero supply but material energy.
Marhaba	C	0.00 hrs	0.00 MW	349.69 MWh	Zero supply and zero load but positive energy.
Lautai	C	0.00 hrs	0.00 MW	237.10 MWh	Zero supply and zero load but positive energy.
Panshekara	C	0.97 hrs	2.54 MW	491.66 MWh	Very low supply but significant energy.
Government House Dutse	A	0.35 hrs	0.05 MW	0.49 MWh	Band A feeder with extremely low reported supply.
Airport Road	B	0.35 hrs	2.80 MW	28.60 MWh	Band B feeder with extremely low supply.

Important note

These issues do not necessarily mean the data is incorrect. However, they should be treated as exception flags that require validation before management decisions are made.

6. Feeder Performance Review

The feeder-level table is one of the strongest parts of the report because it allows management to identify performance differences across feeders. However, the current format is too raw for management use. It should be converted into feeder rankings, exception lists and action categories.

6.1 Stronger-Performing Feeders

Feeder	Band	Avg Supply	Availability	Peak Load	Energy Delivered	Management Interpretation
Industrial	A	23.39 hrs	97.5%	1.50 MW	466.62 MWh	Strong supply availability.
Ceramic	A	23.03 hrs	96.0%	4.80 MW	2,681.55 MWh	Strong supply and high energy delivery.
Funtua Water Works	A	22.55 hrs	94.0%	5.00 MW	191.00 MWh	Strong service delivery.
Yusuf Road	A	22.35 hrs	93.1%	5.00 MW	624.00 MWh	Strong supply performance.
NBC	A	22.29 hrs	92.9%	4.00 MW	502.71 MWh	Strong availability.
Dala Foods	A	22.19 hrs	92.5%	8.00 MW	593.29 MWh	Strong supply performance.
Gwarzo Road	B	21.97 hrs	91.5%	1.70 MW	313.00 MWh	Strong performance despite Band B classification.
Maimalari	A	21.94 hrs	91.4%	5.50 MW	911.00 MWh	Strong availability and energy delivery.
Funtua Textile Mill	A	21.81 hrs	90.9%	23.00 MW	328.40 MWh	Strong supply, with high peak load requiring monitoring.
Bompai	A	21.39 hrs	89.1%	6.80 MW	2,346.00 MWh	Strong supply and high energy delivery.

6.2 Management Interpretation of Strong Feeders

The stronger-performing feeders should be treated as operational benchmarks. Management should assess whether their performance is driven by better upstream supply, lower fault incidence, stronger feeder condition, lower technical constraints, priority loading, or stronger operational control.

The performance of feeders such as Industrial, Ceramic, NBC, Dala Foods, Maimalari and Bompai suggests that the network has pockets of strong service performance. The next step is to understand whether the operational conditions supporting these feeders can be replicated across weaker feeders.

6.3 Weak and Exception Feeders

Feeder	Band	Avg Supply	Availability	Peak Load	Energy Delivered	Management Concern
Hausawa	B	0.00 hrs	0.0%	0.00 MW	533.91 MWh	Data exception requiring validation.
Kadarko	B	0.00 hrs	0.0%	0.00 MW	0.00 MWh	No reported supply or energy.
Karkasara	B	0.00 hrs	0.0%	0.00 MW	0.00 MWh	No reported supply or energy.
Low Cost	B	0.00 hrs	0.0%	0.00 MW	253.78 MWh	Data exception requiring validation.
New Site	B	0.00 hrs	0.0%	0.00 MW	0.00 MWh	No reported supply or energy.
Marhaba	C	0.00 hrs	0.0%	0.00 MW	349.69 MWh	Data exception requiring validation.
Lautai	C	0.00 hrs	0.0%	0.00 MW	237.10 MWh	Data exception requiring validation.
Nagoggo	C	0.00 hrs	0.0%	0.00 MW	0.00 MWh	No reported supply or energy.
Tiga	C	0.00 hrs	0.0%	0.00 MW	0.00 MWh	No reported supply or energy.
Sani Abacha Way	B	0.06 hrs	0.2%	0.75 MW	252.46 MWh	Near-zero supply but material energy.

6.4 Management Interpretation of Weak Feeders

The weak feeder list should be separated into two categories.

The first category is operational underperformance, where feeders show low supply, low availability and low delivered energy. These feeders may require technical intervention, supply restoration, operational review or service-band reclassification.

The second category is data exception, where the feeder shows an inconsistent relationship between supply hours, load and energy delivered. These feeders should be reviewed by the data/performance monitoring team before being used for performance conclusions.

Management should not treat all weak feeders the same. Some require technical recovery, while others require data validation.

7. State Performance Review

The state-level performance summary shows that Kano accounts for the largest feeder base and the highest number of interruptions. Katsina recorded the highest average hours of supply, while Jigawa performed slightly above Kano in average supply.

State	Feeders	Avg Supply	Availability	Interruptions	Peak Load	Management Interpretation
Jigawa	16	7.11 hrs	29.6%	351	9.30 MW	Moderate supply performance, lower interruption burden due to smaller feeder base.
Kano	85	6.95 hrs	29.0%	4,669	22.00 MW	Largest network exposure and highest interruption count. Requires focused operational attention.
Katsina	24	8.77 hrs	36.5%	763	47.00 MW	Best average supply among the three states, but high peak load concentration should be monitored.

7.1 State-Level Management Interpretation

Kano should be the primary focus for operational performance improvement because it accounts for 85 of the 125 monitored feeders and 4,669 interruptions. However, because Kano has the largest feeder base, management should not rely only on absolute interruption count. The report should also calculate interruption intensity per feeder.

Using the reported figures:

State	Interruptions	Feeders	Interruptions per Feeder	Interpretation
Jigawa	351	16	21.9	Lowest interruption intensity.
Kano	4,669	85	54.9	Highest interruption intensity and highest operational concern.
Katsina	763	24	31.8	Moderate interruption intensity.

State-level conclusion

Kano is not only the largest state by feeder count, but also has the highest interruption intensity per feeder. This strengthens the case for a focused Kano reliability improvement plan.

8. Service Band Performance and Compliance Review

The service-band summary shows a significant mismatch between service-band expectations and actual performance for several feeders. Band A is the only band with relatively strong average supply. Bands B, C and D are materially weak.

Service Band	Feeders	Avg Supply	Interruptions	Management Interpretation

Service Band	Feeders	Avg Supply	Interruptions	Management Interpretation
Band A	21	19.83 hrs	656	Relatively strong, but some Band A feeders still require exception review.
Band B	39	5.98 hrs	2,382	Very weak for Band B and requires urgent review.
Band C	44	4.49 hrs	1,879	Weak performance, with many feeders below expected supply levels.
Band D	20	3.05 hrs	732	Very poor service availability.
Band E	1	6.94 hrs	134	One feeder only, but interruptions are high relative to feeder count.
Total	125	7.32 hrs	5,783	Overall service delivery remains weak.

8.1 Service-Band Management Interpretation

Band A performance is the strongest, but it still requires detailed review because some Band A feeders recorded weak performance. For example, Government House Dutse recorded only 0.35 hours average supply despite being classified as Band A. This kind of exception should be separately reviewed because it may indicate either a data issue, a prolonged outage, or a feeder classification issue.

Band B is the most important management concern. It has 39 feeders, recorded 2,382 interruptions, and averaged only 5.98 hours of supply. This is weak for the band and indicates a potential service-band delivery problem.

Band C has the largest number of feeders, with 44 feeders, and averaged only 4.49 hours. This suggests that a large share of the monitored network is operating at a low service level.

Band D averaged 3.05 hours, which indicates very poor service availability. Band E recorded 6.94 hours, but this is based on only one feeder, Mai Ruwa, and should not be overinterpreted as a broad band-level trend.

8.2 Recommended Service-Band Recovery Actions

Band	Key Issue	Recommended Management Action
Band A	Mostly strong, but with outlier feeders showing very poor supply.	Investigate Band A outliers and separate genuine service failure from data issues.
Band B	Average supply is materially weak at 5.98 hours.	Develop feeder-by-feeder Band B recovery plan, starting with high-customer and high-energy feeders.
Band C	Largest feeder count and weak average supply.	Prioritise feeders with high interruption duration and high commercial value.
Band D	Very low supply at 3.05 hours.	Review technical constraints and determine whether improvement or reclassification is required.
Band E	One feeder only, but with high interruption count.	Monitor Mai Ruwa separately and confirm whether its performance reflects actual service conditions.

9. Priority Management Issues

Priority Issue	Evidence from Report	Management Risk	Required Response
Weak overall supply level	Average supply is 7.32 hours/day.	Customers remain underserved despite improvement.	Maintain supply improvement momentum and target weak feeders.
High interruption burden	5,781 to 5,783 interruptions reported.	Reliability remains weak and customer complaints may rise.	Implement interruption reduction plan.

Priority Issue	Evidence from Report	Management Risk	Required Response
Long interruption duration	Average interruption duration is 16.68 hours.	Outages are prolonged and restoration may be slow.	Track restoration performance by feeder and fault type.
Large N/A interruption category	2,768 N/A interruptions.	Root-cause analysis is weak.	Clean up interruption coding and enforce cause classification.
Band B underperformance	Band B averages only 5.98 hours.	Potential service-band and customer expectation risk.	Launch Band B feeder recovery plan.
Data anomalies	Some feeders show zero supply but positive energy.	Management decisions may be based on unvalidated data.	Create data exception register.
Kano interruption intensity	Kano records 4,669 interruptions and 54.9 interruptions per feeder.	Kano is the main reliability hotspot.	Develop Kano-specific reliability intervention plan.

10. Recommended Management Action Plan

Action Area	Recommended Action	Responsible Team	Timeline	Expected Output
Data reconciliation	Reconcile interruption totals across technical overview, interruption breakdown, state summary and service-band summary.	Performance Monitoring / Data Team	Before next report	One validated interruption figure across all sections.
Energy methodology	Clarify the basis for total energy and daily average energy calculations.	Data Team / Metering Team	Before next report	Methodology note included in report.
Interruption classification	Review and reclassify N/A interruption events.	Network Operations / Control Room	2 weeks	Reduced N/A share and better root-cause profile.
Load-supply analysis	Break down L/S into clearer subcategories.	Network Operations	1 month	Better understanding of supply constraint drivers.
Local fault response	Track each local fault from report time to restoration time.	Technical Operations / Fault Response	Monthly	Reduced local fault turnaround time.
Weak feeder review	Create bottom-feeder list for management review every month.	Regional Operations	Monthly	Targeted feeder recovery plan.
Data exception register	Flag feeders with zero supply but positive energy, zero load but positive energy, or abnormal peak load.	Performance Monitoring / Metering	Monthly	Cleaner feeder performance data.
Band B recovery	Identify Band B feeders below threshold and prepare recovery actions.	Commercial / Technical / Regional Operations	1 month	Band B service recovery tracker.
Kano reliability plan	Prioritise Kano feeders with highest interruption count and duration.	Kano Regional Operations	1 month	Kano feeder reliability intervention plan.
Management accountability	Add action owners, deadlines and status tracking to the monthly report.	Executive Management / PMO	Immediate	Management report becomes an accountability tool.

11. Proposed Management Report Format Going Forward

The current report is visually strong and useful as a monitoring dashboard. To make it a full management report, the format should be expanded slightly without making it too narrative-heavy.

The proposed structure for future monthly reports is:

Section	Purpose	Content
Executive Management Summary	Gives management the main message quickly.	Key performance highlights, risks and priority actions.

Section	Purpose	Content
KPI Dashboard	Shows overall performance.	Current month, previous month, movement, target and status.
Data Quality Note	Protects the credibility of the report.	Definitions, assumptions, missing data, estimated data and validation issues.
Reliability Review	Explains interruption performance.	Interruption count, duration, top causes and restoration performance.
Feeder Performance Review	Highlights strong and weak feeders.	Top performers, bottom performers and exception feeders.
State Performance Review	Supports regional accountability.	State averages, interruption intensity and regional actions.
Service Band Review	Links performance to service expectations.	Actual supply by band, variance, weak feeders and recovery plan.
Management Action Tracker	Converts insight into action.	Issue, owner, action, timeline and status.
Appendix	Keeps the main report clean.	Full feeder table and raw data exceptions.

12. Suggested Executive Dashboard Page

For management use, the first page after the cover should include the following dashboard.

Metric	May 2026	Previous Period	Movement	Status
Average supply	7.32 hrs	5.01 hrs	+46.2%	Amber
Average load	0.54 MW	0.55 MW	-2.1%	Amber
Energy delivered	57,029.86 MWh	Not directly comparable	Improved based on trend	Amber
Total interruptions	5,781 / 5,783	Not stated	Requires reconciliation	Red
Cumulative interruption hours	64,635 hrs	Not stated	Not stated	Red
Average interruption duration	16.68 hrs	Not stated	Not stated	Red
Local fault turnaround time	7.26 hrs	Not stated	Not stated	Red
Band A average supply	19.83 hrs	Not stated	Not stated	Green
Band B average supply	5.98 hrs	Not stated	Not stated	Red
Band C average supply	4.49 hrs	Not stated	Not stated	Red
Band D average supply	3.05 hrs	Not stated	Not stated	Red

13. Suggested Management Commentary for the Dashboard

The May 2026 performance shows improved energy delivery and higher average hours of supply relative to the previous reporting period. However, the overall service delivery position remains weak, with average supply of only 7.32 hours per day across the monitored feeder base.

The improvement in energy delivered appears to have been driven primarily by improved supply availability rather than increased average load, as average load declined slightly during the period. This suggests that management should focus on sustaining supply availability while also investigating whether weak loading reflects suppressed demand, feeder constraints, customer behaviour, commercial issues or data quality limitations.

Reliability remains the most significant concern. The system recorded high interruption exposure, long average interruption duration and a large number of interruption events. The large N/A category in the interruption breakdown also limits the ability to identify root causes and should be treated as an immediate data governance issue.

Service-band performance is also a major concern. Band A performed relatively well on average, but Bands B, C and D recorded weak supply levels. Band B requires particular attention because it has 39 feeders, recorded 2,382 interruptions, and averaged only 5.98 hours of supply. This level of performance should trigger a feeder-level service recovery review.

Management attention in the next reporting cycle should focus on interruption reduction, data validation, feeder exception management, and service-band recovery.

14. Conclusion

The May 2026 11kV performance report shows that KEDCO made progress in improving hours of supply and energy delivery. However, the company's overall operational performance remains constrained by high interruptions, long outage durations, weak service-band delivery and data-quality gaps.

The most urgent issue is not only the level of interruptions, but the quality of interruption classification. With N/A representing the largest interruption category, management does not yet have a sufficiently clear root-cause view of system reliability. This should be addressed immediately.

The second major issue is service-band underperformance. While Band A is relatively strong on average, Bands B, C and D are materially weak and require focused feeder-level recovery plans.

The third issue is data validation. Feeders with zero supply but positive energy should be flagged and reviewed before management conclusions are drawn. This will strengthen confidence in the tool and make the report more useful for operational decision-making.

Overall, the report has a strong foundation and can become a proper management tool if it moves beyond presenting data and consistently provides interpretation, exception flags, accountability and action tracking.

15. Immediate Next Steps

Step	Action	Output
1	Reconcile interruption totals across the report.	Clean validated KPI set.
2	Review and reclassify N/A interruptions.	Improved root-cause diagnosis.
3	Validate feeders with zero supply but positive energy.	Data exception register.
4	Prepare Band B, C and D recovery lists.	Service-band improvement plan.
5	Add action owners and timelines to the next report.	Management accountability tracker.
6	Add a methodology note for energy, load, supply hours and interruption classification.	Stronger report credibility.

Concluding position

The report should retain its current visual dashboard style, but the management version should add interpretation, exception reporting and action tracking so that it becomes a decision-making document rather than only a monitoring output.